BRTH R2 — physical QR scan test sheet

TEST ONLY — NOT A SECRET — DO NOT USE AS A WALLET.
Every code on this sheet encodes a public, deterministic, 32-zero-byte TEST seed. The identity is a published test vector. Never treat it as a credential, a wallet or a real identity, and never reuse the zero seed for anything real.

Expected TEST IdentityID: `b1:n4xq6w26e76t5acvynw7dsf6c53pm6nc2detgqihszkcjqkvweja`

Physical scanning is UNTESTED. No printer, scanner or camera was available on the machine that produced this sheet, so the project has performed *no* physical print/camera/phone scan. This sheet is the opt-in physical test. Every physical claim remains a hypothesis until a volunteer reports back.

A. PRIVATE BACKUP TEST QR (BRQ1S)

Custody class: **PRIVATE BACKUP SECRET**

0.4 mm per module → prints 14.8 × 14.8 mm



payload 37 bytes
SHA-256 `857e9e2bddf6bed2afbb7f598f6c0df6dacd361226505d07f846942eefad266e`
QR v3, EC L, 29 modules/side

0.5 mm per module → prints 18.5 × 18.5 mm

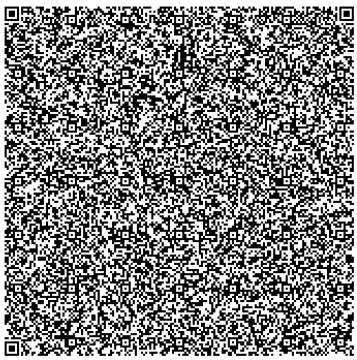


payload 37 bytes
SHA-256 `857e9e2bddf6bed2afbb7f598f6c0df6dacd361226505d07f846942eefad266e`
QR v3, EC L, 29 modules/side

B. PUBLIC IDENTITY DESCRIPTOR TEST QR (BRQ1D)

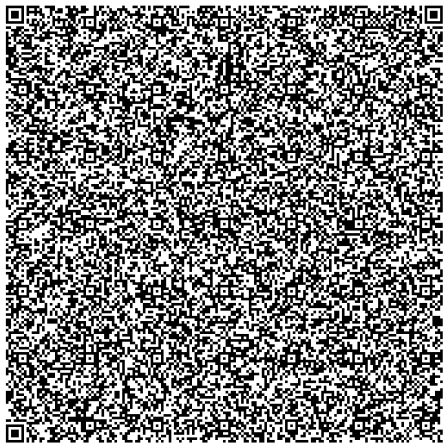
Custody class: **PUBLIC VIEWER IDENTITY DESCRIPTOR**

0.4 mm per module → prints 70.8 × 70.8 mm



payload 2693 bytes
SHA-256 `5b52c6a6be822f047e82cf62764966016c8d5c22c9fa3c54825d1173ed27b91c`
QR v38, EC L, 169 modules/side

0.5 mm per module → prints 88.5 × 88.5 mm



payload 2693 bytes
SHA-256 `5b52c6a6be822f047e82cf62764966016c8d5c22c9fa3c54825d1173ed27b91c`
QR v38, EC L, 169 modules/side

How to run this test (volunteer with a phone)

1. Print this sheet at **100% / actual size** — do not use “fit to page” or “shrink”, which changes the module size and invalidates the test. Print every page of the sheet.
2. Scan with your phone's stock camera app. Note the app name and version.
3. For code B (BRQ1D, 2693 raw binary bytes) you need a scanner app that can return **raw binary bytes**. A text-only scanner will corrupt or refuse this payload — that is a tooling limitation, not a QR failure.
4. Record for each code: app name, distance (cm), angle, lighting condition, and success / failure.
5. Compare the recovered bytes' SHA-256 against the value printed next to the code. If your app can only export text, export hex and use the `--bytes-hex` option of the offline verifier.

Offline verification

Fully offline; no network, no Punkti node, no DNS:

```
python3 scan_verify.py --self-test          # against the local test artifacts
python3 scan_verify.py scan.png                      # a photo of the printed sheet
python3 scan_verify.py scanned_bytes.hex           # raw bytes exported by your app
```

A pass means: the decoded bytes match the printed SHA-256, and the payload reconstructs the expected TEST IdentityID with the unchanged Layer-4 PQ code. It does **not** mean the physical design is validated for real identities, and for code B it does not prove possession or current authority.

Code	Custody	Proves	Does NOT prove
A — BRQ1S	PRIVATE BACKUP SECRET (never publish)	Offline recovery of the root identity from the QR alone	Current authority, legal identity, physical authenticity, 100-year algorithm survival
B — BRQ1D	PUBLIC VIEWER IDENTITY DESCRIPTOR (displayable)	The canonical Birth, its digest and the IdentityID are internally consistent	Proof of possession, current authority, revocation status, freshness, intended actor, external authenticity